DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS: X | **FULL MARKS: 40**
SUBJECT: CHEMISTRY | **DATE: 21/10/2024**

General Instructions:

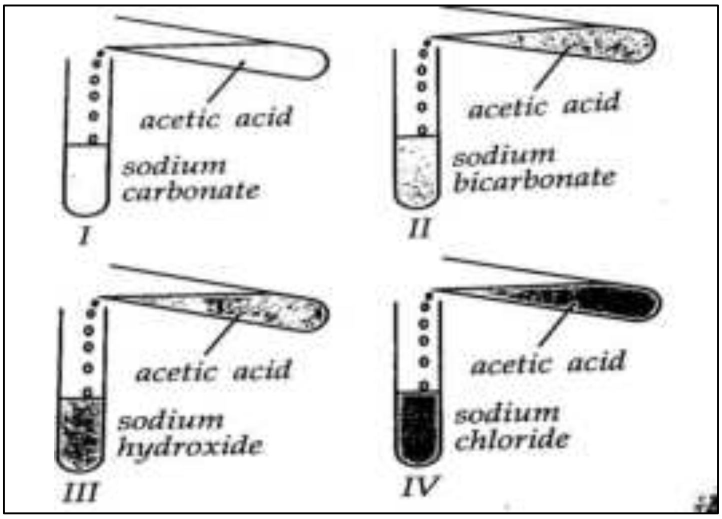
- **The paper consists of four printed pages.**
- **Read the questions very carefully.**
- **Answers should be to the point.**
- **Question numbers should be copied carefully while answering the questions.**

SECTION A
(*Attempt all questions*)

Question:1

Choose the correct answers from the options given below: **[8]**

a) Roshan added acetic acid to test tubes I, II, III and IV and then introduced a burning candle near the mouth of each test tube.



The candle would not be extinguished near the mouth of test tubes:
(A) I and II
(B) II and III
(C) III and IV
(D) I and IV

b) The IUPAC name of the product of the reaction of boiling ethyl bromide with alcoholic caustic potash is:
(A) Ethanol
(B) Ethylene
(C) Ethene
(D) Acetylene

c) Assertion: The volume of 4.4 g of carbon dioxide and 6.4 g of sulphur dioxide is equal at STP.
Reason: The number of moles is equal in both cases
(A) Both A and R are true, and R is the correct explanation of A.
(B) Both A and R are true, and R is not the correct explanation of A.
(C) A is false but R is true.
(D) A is true but R is false.

d) The ratio between the number of molecules in 2 g of hydrogen and 32 g of oxygen is:
   (A) 1:2
   (B) 1: 0.01
   (C) 1:1
   (D) 0.01: 1

e) $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2H_2O + 2X\uparrow$
   The gas X turns:
   (A) Lime water milky.
   (B) Gives white precipitate when bubbled through lead acetate solution.
   (C) Turns moist potassium iodide paper brown.
   (D) Gives a yellow precipitate when bubbled through silver nitrate solution.

f) The gas which turns nitric acid yellow is:
   (A) Sulphur dioxide
   (B) Nitrogen dioxide
   (C) Carbon dioxide
   (D) Hydrogen sulphide

g) In the equation given below the role played by sulphuric acid is:
   $S + 2H_2SO_4 \text{ (conc)} \rightarrow 3SO_2 + 3SO_2$
   (A) Non volatile acid
   (B) Oxidizing agent
   (C) Dehydrating agent.
   (D) Typical acid

h) Which of the following metal can be used to distinguish between dilute and concentrated sulphuric acid:
   (A) Copper
   (B) Magnesium
   (C) Iron
   (D) Calcium

Question:2

a) Give reasons for the following: [5]
   (i) Bromine water cannot be used to distinguish between ethylene and acetylene.
   (ii) Sulphuric acid is not obtained by dissolving sulphur trioxide in water.
   (iii) Burning of methane in limited supply of air is dangerous.
   (iv) Nitric acid cannot be concentrated beyond 68%
   (v) Vanadium pentoxide is preferred over platinum as a catalyst for contact process.

b) What would you observe when: [5]
   (i) A beaker containing concentrated sulphuric acid is left open in the air.
   (ii) Dilute nitric acid is reacted with copper turning.
   (iii) Acetylene is burnt with excess air.
   (iv) Acetic acid is cooled to a temperature of 16.5°C
   (v) Acetic acid is reacted with ethanol in presence of concentrated sulphuric acid.

Question;3

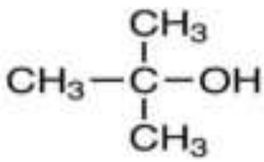
Draw the structure of the following: [2]
 a) 2-methylpropanoic acid
 b) 2,2,2- trichloroethanal

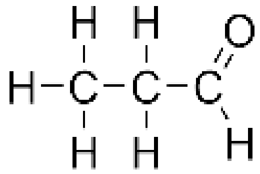
SECTION B

(*Attempt all questions*)

Question: 4

a) **Write the IUPAC name of the following compound:**

(i) 

(ii)  **[2]**

b) **Write balanced chemical as example of following reactions:**

- **(i) Decarboxylation reaction**
- **(ii) Dehydrohalogenation reaction**
- **(iii) Substitution reaction.** **[3]**

Question:5

a) **While performing an experiment to determine the relative molecular mass and formula of a gaseous fluoride of phosphorus, Richard observed 112 cc of the gaseous fluoride of phosphorus has a mass of 0.63 g at STP. Calculate the relative molecular mass of the fluoride. If the molecule of fluoride contains one atom of phosphorus, then determine the molecular formula of fluoride of phosphorus.**
[P = 31 F = 19]. **[2]**

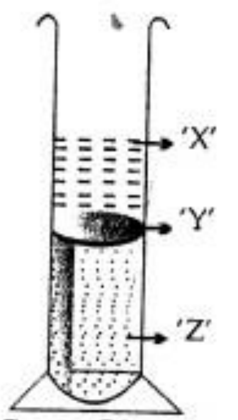
b) **A gas cylinder filled with hydrogen holds 5 g of the gas. The same cylinder holds 85 g gas X under the same conditions of temperature and pressure. Calculate:**

- **(i) Vapour density of gas X.**
- **(ii) Molecular mass of gas X.** **[2]**

c) **State Lussac's law of combining volume.** **[1]**

Question:6

a) **The familiar brown ring test for nitrates depends on the ability of Fe^{2+} to reduce nitrates to nitric oxide, which reacts with Fe^{2+} to form a brown coloured complex. The test is usually carried out by adding dilute ferrous sulphate solution to an aqueous solution containing nitrate ion, and then carefully adding concentrated sulphuric acid along the sides of the test tube. Result of the test is given below:**



Brown Ring Test

- **(i) Identify 'X' and 'Z'.**
- **(ii) Write the chemical name of the brown ring (Y).** **[3]**

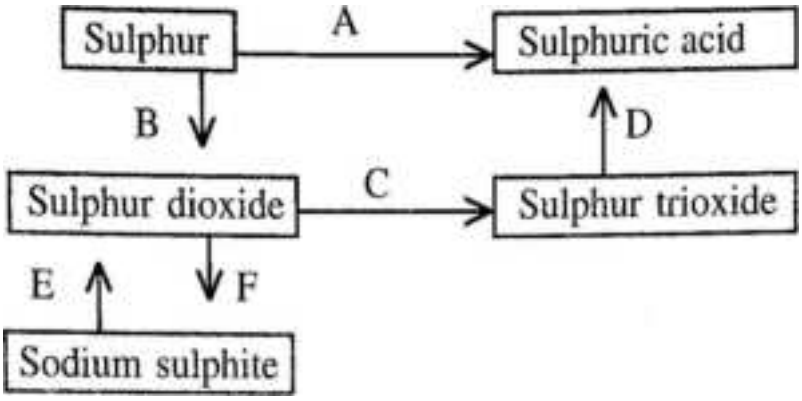
b) In the year 1902 Wilhelm Ostwald developed the method of industrial preparation of nitric acid. The process is highly energy – efficient and cost-effective answer the following with respect to this process:

(i) Write equation for the catalyzed reaction.

(ii) Explain why excess of air is taken for this process. [2]

Question:7

Study the diagram given below and answer the following questions:



(i) Write balanced equation for conversion A.

(ii) State the conditions for the conversion C.

(iii) Write balanced chemical equations for conversion D.

(iv) What type of substance is added to sodium sulphite for conversion. [5]